S6 B Tech - CS4089E Machine Learning Lab
Winter 2025-2026

Experiment 8 - Upload by 02/04/2026, 11 PM
Convolutional Neural Network

1: Create a Convolutional Neural Network (CNN) model to predict a digit value from an image representing a digit. Build a web application using Streamlit to upload an image, display the image uploaded and show the predicted digit (using the CNN model built).

The image dataset is to be created as given below:

Create images of size 1x10 corresponding to the digits (0 to 9). Assume that all white cells in an image represent the digit 0, and all black cells represent the digit 9. Digit 1 is represented with any one of the cells black, digit 2 is represented with any two of the cells black and a similar pattern for digits from 3 to 8.

Create a dataset of 800 random images of such representations, covering all the 10 digits. Form an 80-20 training set and test set split of the dataset. Properly address class imbalance issues if any.

Create your own functions for: Convolution, Pooling, Padding and Fully Connected Layers. Provide options to set the values of size of kernels, stride values by the user.